

High-fidelity facial reconstruction from a single photo using photo-realistic rendering

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Figure 1: 3D face reconstructions generated from single frontal photos using a convolutional neural network to estimate the initial head shape followed by landmark-based deformation for optimal likeness.

ABSTRACT

We propose a fully automated method for realistic 3D face reconstruction from a single frontal photo that produces a high-resolution head mesh and a diffuse map. The photo is input to a convolutional neural network that estimates the weights of a morphable model to produce an initial head shape that is further adjusted through landmark-guided deformation. Two key features of the method are: 1) the network is exclusively trained on synthetic photos that are photo-realistic enough to learn real shape predictive features, making it unnecessary to train with real facial photos and corresponding 3D scans; 2) the landmarking statistical errors are incorporated in the reconstruction for optimal accuracy. While the method is based on a limited amount of real data, we show that it robustly and quickly performs plausible face reconstructions from real photos.

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1 INTRODUCTION

Customized and engaging virtual experiences are sought-after by many industries, including augmented and virtual reality, gaming, fashion, and entertainment. To see oneself represented faithfully in a virtual world is crucial to the success of those experiences. Standard procedures to generate realistic 3D face avatars typically require a combination of expensive head capturing systems (e.g. light stages) and extensive manual work from skilled artists. Automating the 3D reconstruction of the human face is an active research area, pioneered by [Blaiz and Vetter 1999] with the introduction of 3D morphable models (3DMM). The scarceness of facial datasets with 3D ground truth shapes has been a hindering issue.

We developed a convolutional neural network (CNN) to directly estimate the shape coefficients of a custom 3DMM from a single frontal photo with neutral expression. The CNN was trained in a supervised setting using reasonably photo-realistic synthetic photos. The predicted mesh is refined by landmark-guided deformation to adjust for subtle facial features not captured by the 3DMM, and to transform the input photo into UV space in order to produce a diffuse map.

2 METHOD

The proposed method is illustrated in Figure 2. We built a custom 3DMM by combining the ICT Face Model Light [Li et al. 2020], which consists of 100 head shape identities, with 50 additional head shape models from both females and males purchased from 3D Scan Store (<https://www.3dscanstore.com>) and Triplegangers (<https://triplegangers.com>). This involved porting all meshes to the same topology using an in-house non-rigid iterative closest point

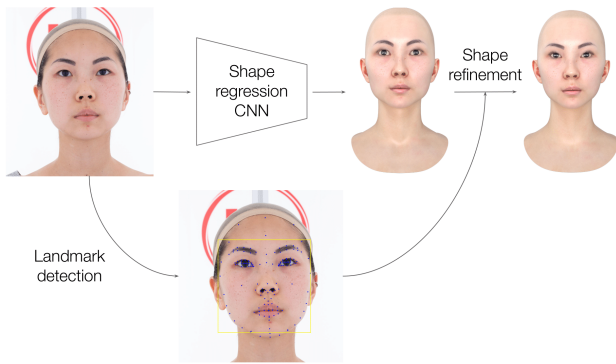


Figure 2: Method overview. We generate varied head shape-render pairs to train a shape regression CNN. During reconstruction, the head shape predicted by the CNN is refined by landmark-based deformation.

algorithm implementation, and performing principal component analysis to produce a 3DMM with 150 meaningful mesh identities that captures a wide range of human head shapes.

In order to train our shape regression CNN, we devised a method to simulate a number of head shape / photo pairs by sampling the 3DMM, and rendering the resulting head meshes in a photo-realistic manner.

For rendering, we randomly selected albedo textures for the face and eyes from comprehensive databases. Occlusions by hair, beard, and hats were optionally simulated. The focal length and sensor size of the camera were also randomized based on the camera settings of four smartphones – Apple iPhone 11, Xiaomi Mi Note 10, Google Pixel 5, and HUAWEI P40 Pro. For lighting and background, we used high dynamic range (HDR) images covering varied conditions (interior/exterior, bright/low light, and day/night). The head rotation and camera position were also randomized within pre-defined ranges. The renders were produced with Blender (<https://www.blender.org>) using Cycles as the render engine and custom materials for all assets. Each head shape was rendered twice under different cameras and HDR images, so that the CNN can learn to reconstruct the same shape consistently under varied perspectives.

The shape regression CNN was trained using the Densenet 121 architecture [Huang et al. 2017] on 15000 synthetic head shape-photo pairs to predict head shape from a photo. This was done by minimizing the mean squared error (MSE) between the 3DMM coefficients output by the CNN and the ones that served to simulate the photo. The CNN weights were initialized from image classification training with ImageNet [Deng et al. 2009], which was retrospectively found to decrease the average MSE on the validation data. While we do not match the realism of recent data simulation pipelines [Wood et al. 2021], the quality reached proved to be good enough to learn features that describe the face shape in real photos, making it unnecessary to train the CNN with any real photo-3D scan pairs or pre-train it on a related task (e.g. facial recognition).

The 3DMM-based head shape estimated by the CNN is refined in a deformation process that uses 140 facial landmarks defined by us to outline the eyes, nose, mouth, jawline, cheekbone and forehead (see Figure 2). We trained a cascade of regressions trees [Kazemi and

Sullivan 2014] to detect these landmarks on the input photo, and devised a mesh deformation method that minimizes the mean 2D squared Euclidean distance between the detected photo landmarks and the ones projected onto the image plane from the 3D shape. Our method differs from the one proposed by [Wang et al. 2019] in that it weights the landmarks according to their standard statistical errors estimated in a validation dataset; hence, eye landmarks tend to have more influence on the final shape than jawline landmarks, for instance, as they are typically more accurate.

3 RESULTS AND CONCLUSION

We performed reconstruction from photos of 6 subjects – Caucasian, Asian, and African males and females - for which we have 3D ground truth face meshes. The vertex-wise Euclidean reconstruction errors are sub-centimetric, averaging between 0.314 cm and 0.419 cm.

Additionally, we used three off-the-shelf automated face recognition services from AWS (<https://docs.aws.amazon.com/rekognition/latest/dg/faces-comparefaces.html>), Azure (<https://azure.microsoft.com/en-us/services/cognitive-services/face/>), and Face ++ (<https://www.faceplusplus.com/face-comparing/>), to compute similarity scores between the input photo and face renders performed before and after the deformation step, respectively, for a random sample of 100 photos from Caucasian, Asian, African and mixed-race subjects with balanced gender and age (in the range 12–60 years). The distribution of similarity scores suggests that our facial reconstruction generally achieves good likeness, with median scores of 97.58% for AWS, 61.33% for Azure, and 85.64% for Face++. Despite the inherent discrepancies between scoring methods, one-sided Wilcoxon tests on the score differences before and after shape refinement showed that all of them (AWS, $p = 1.79 \times 10^{-9}$; Azure, $p = 1.48 \times 10^{-10}$; Face++, $p = 2.52 \times 10^{-5}$) picked a significant increase in similarity due to deformation.

Figure 1 shows the reconstruction renders of two input photos, subjectively confirming the possibility to accurately reconstruct the facial shape from a single frontal photo. We achieve similar realism across real photos from a wide range of human faces with a computation time of the order of 10 seconds on CPU for shape reconstruction, making our method well-suited to streamline the creation of lifelike avatars at scale. Future work includes enhancing the 3DMM using additional head shapes to cover an even wider range of human faces, as well as facial expressions to enable input photos with non-neutral expressions.

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